18-01-2025

Agenda :-

→ Yolo history

→ Yolo v1 :

→ Yolo v5 implementation :-

Yolo : You only look once

→ One stage detector

Yolo →

v1 → 2016

→ Alexey Bochkovskiy, Ali farhadi, Ross Girshick, Joseph redmond

V2 → 2017

V3 → 2018

V4 → 2020

Alexey Bochkovskiy

V6 → Meituan Yolo → 2022

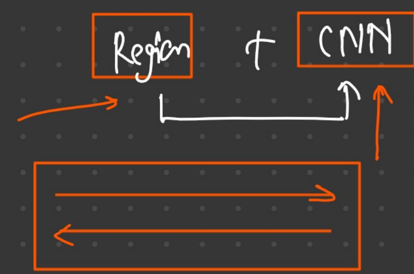
V7 → Alexey

Two stage Detectors

→ RCNN

→ Fast RCNN

→ Faster RCNN



→ 1 fps, 4 fps

→ 30 fps → 1 fps

29 fps

Timeline <u>Yolo ultralytics</u>

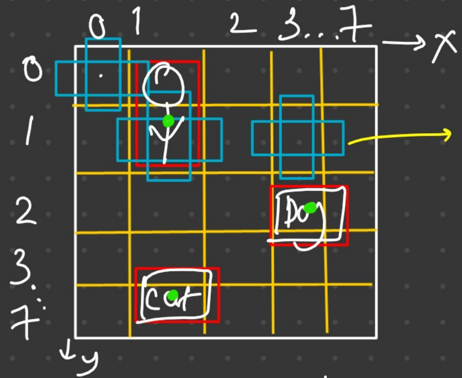
Yolo v5 – May 2020

Yolo v8 → Jan 2023

$\boxed{\begin{array}{l} \text{v9} \to 2024 \\ \text{v10} \to 2024 \\ \text{v11} \to 2024 \end{array}}$

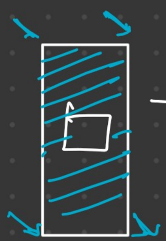
<u>Yolo v1</u>

Image → Divide the image into grids → For each grid they have anchor boxes → Each anchor box prediction → filtering of predicted boxes → NMS → <u>o/p</u>

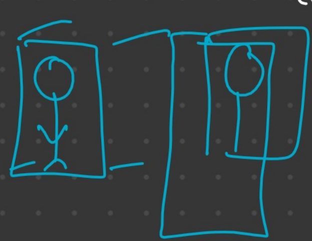


(1,4) (1,9)

1. Divide the i/p image into grids. 7x7 → 49
2. For each grid we will assign anchor boxes. 2
3. Each anchor box is responsible for prediction. [x, y, w, h, conf_obj]



[x, y, w, h, object] → confidence



1. Dataset → $\underline{\underline{\text{20 labels}}}$

7x7 — 49

↓

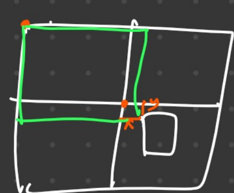
2 anchor

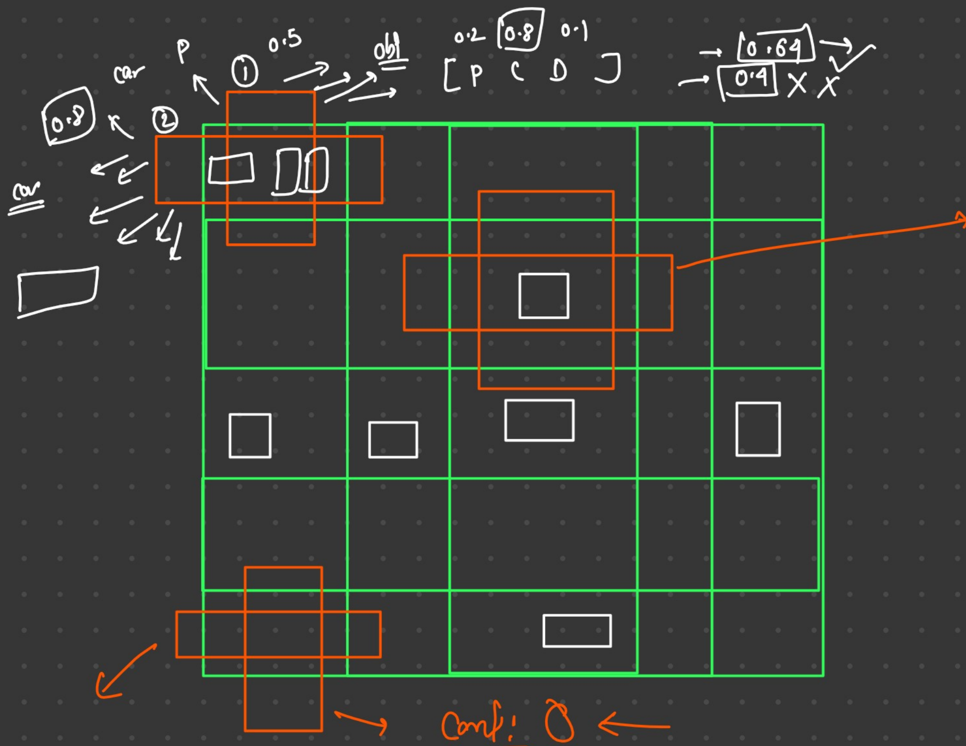
↓

[x, y, w, h, object]

[class probabilities] → [0 .... 19]

$[\underbrace{5}_{A1} + \underbrace{5}_{A2} + 20]$ $\underline{\underline{\text{30 values}}}$ x $\boxed{49}$ (7x7)





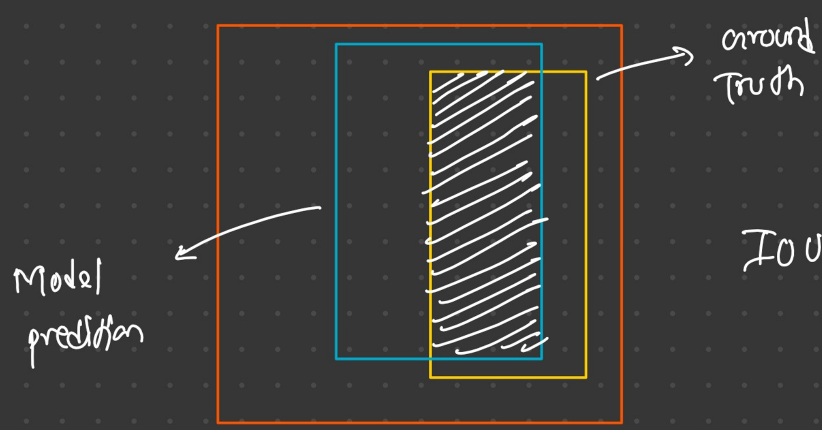
obj -> 0.8
0.9
0.8

conf: 0

x, y, w, h

A1 -> conf: 0

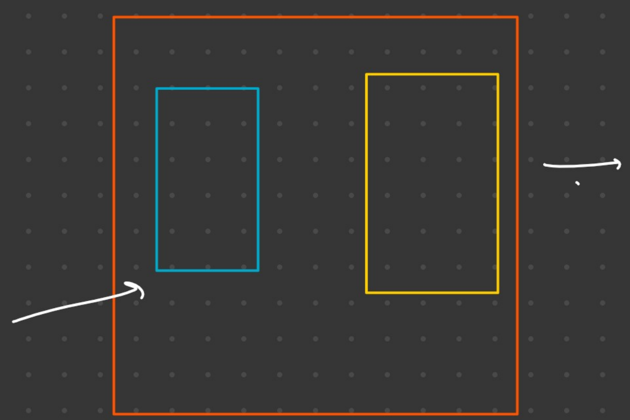
A2 -> conf: 0



$\boxed{IOU > 0.2}$

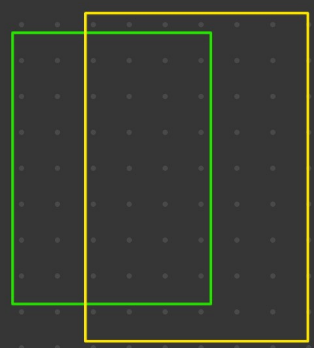
$IOU = \dfrac{70}{\text{total region}}$

=



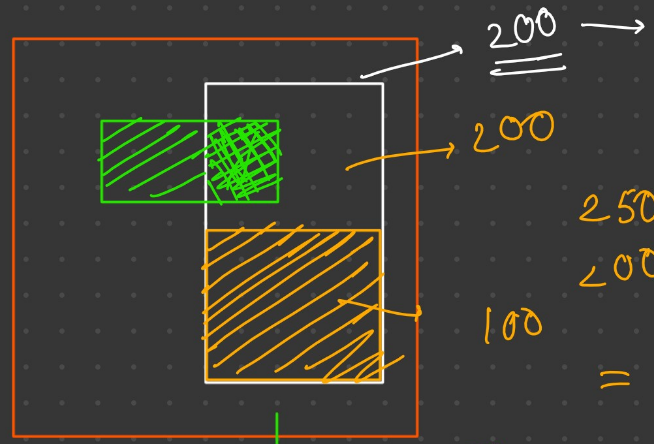
IOU = None

-> 0



$\boxed{IOU > 0.5}$

IOU > 0.9



$\boxed{\frac{200+80}{\boxed{280}}} =$

GT+P

250

200

$= \frac{\text{Intersection}}{\text{Total Area}}$

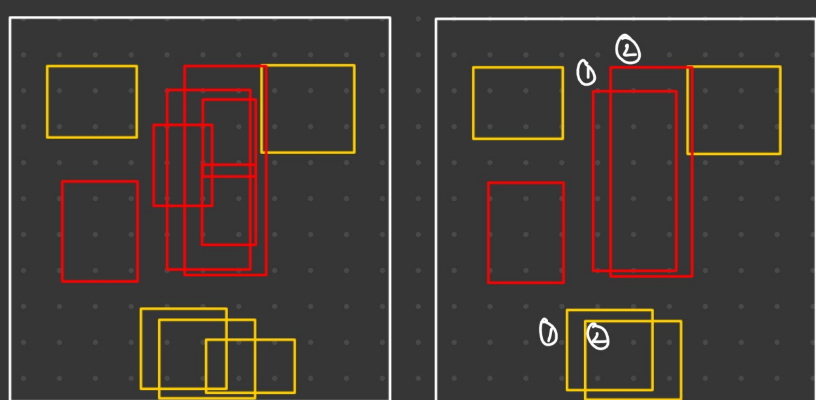
$= \frac{100}{200} = 0.5$

NMS

Non Max Supression

if conf < 0.6

NMS



$$\begin{bmatrix} x_1 & y_1 & w_1 & H_1 & 0.8 & \text{Person} \\ x_1 & y_1 & w_1 & H_1 & 0.78 & \text{CAR} \\ x_1 & y_1 & w_1 & H_1 & 0.9 & \text{Person} \\ x_1 & y_1 & w_1 & H_1 & 0.7 & \text{Person} \end{bmatrix} \rightarrow \text{Sort by Score}$$

$$\begin{bmatrix} x_1 & y_1 & w_1 & H_1 & 0.8 & \text{Person} \\ x_1 & y_1 & w_1 & H_1 & 0.9 & \text{Person} \\ x_1 & y_1 & w_1 & H_1 & 0.7 & \text{Person} \end{bmatrix} \quad \begin{bmatrix} x_1 & y_1 & w_1 & H_1 & 0.78 & \text{CAR} \end{bmatrix}$$

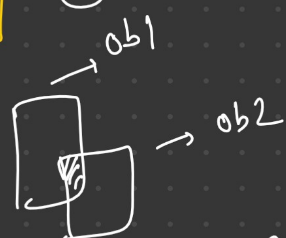
(2) Sort by Score

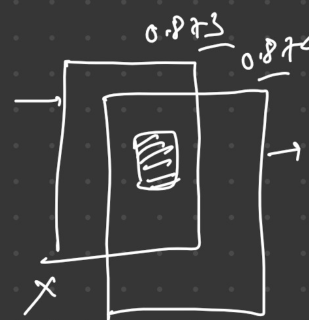
$$\begin{bmatrix} x_1 & y_1 & w_1 & H_1 & 0.9 & \text{Person} \\ x_1 & y_1 & w_1 & H_1 & 0.8 & \text{Person} \\ x_1 & y_1 & w_1 & H_1 & 0.7 & \text{Person} \end{bmatrix}$$

(3) Extract Boxes $\longrightarrow$ (4) Calculate IOU

$$\begin{bmatrix} x_1 & y_1 & w_1 & H_1 \\ x_1 & y_1 & w_1 & H_1 \\ x_1 & y_1 & w_1 & H_1 \end{bmatrix} \begin{matrix} \rightarrow (1) \rightarrow \\ \rightarrow (2) \rightarrow \\ \rightarrow (3) \end{matrix}$$

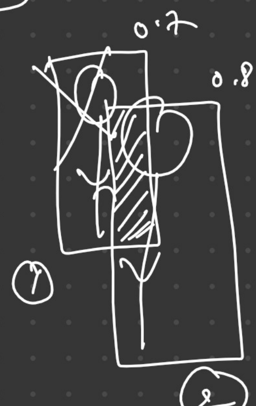
If IOU of (1) & (2) > $\underline{IOU > 0.3}$





→ high IOU →

→ bbox with high probability →



> 0.7 , 0.8 , 0.9

$\underline{\underline{> 0.3}}$

Remove object that have high IOU with the bbox that have the highest probability.

IOU

Person

3

x y w H — 1 → 1

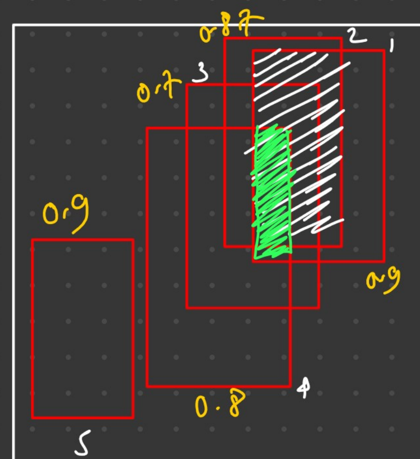
~~x y w H — 2~~ → 2

— 3 → 3

~~x y w H~~

x y w H — 4

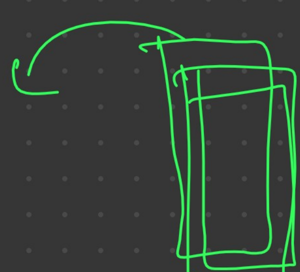
x y w H — 5



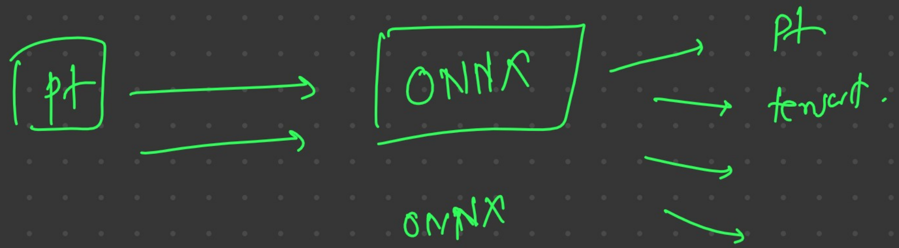
1 ~~2~~ 3

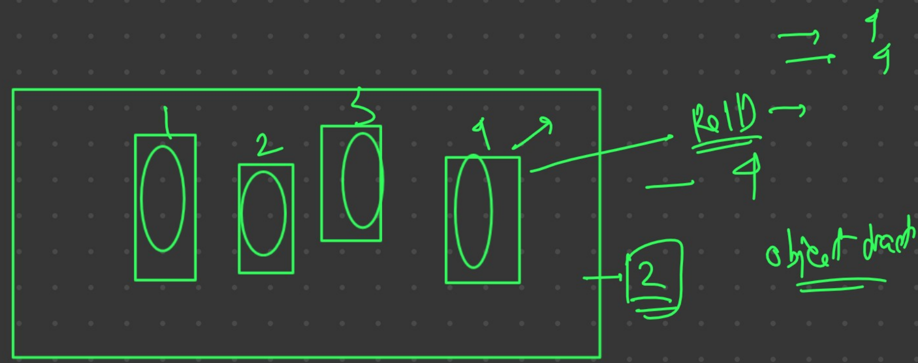
```
def NMS(Box1, Box2):
    if IOU(Box1, Box2) > 0.6:
        if Bo1x.pred_score > Bo1x2.pr:
            remove Box2
        else:
            remove Box1
```

IOU = 0.2 0

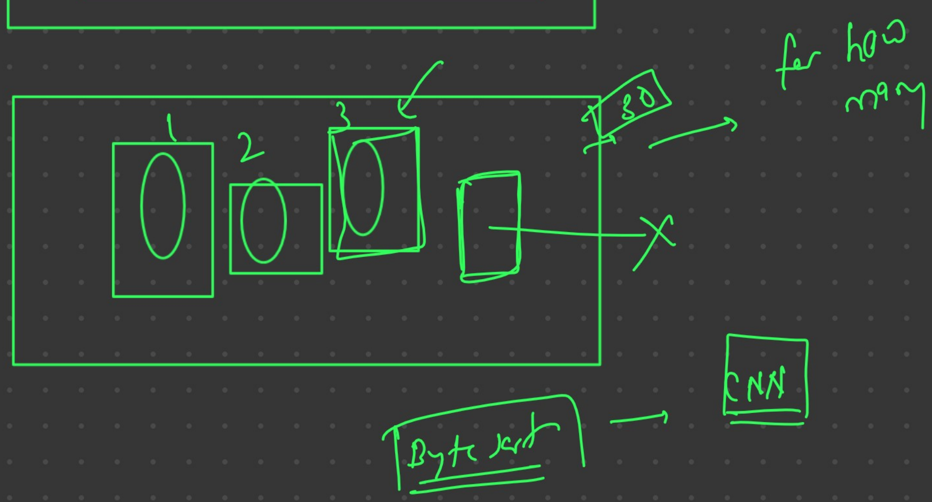


io> 0.9 $\boxed{0.97}$


pt
ONNX
ONNX
pt
tensorrt.


1
2
3
4
ReID
4
4
4
2
object drawn


5
bbox
pasted


1
2
3
80
for how many
Bytetrack
CNN